OpenClaw Full Stack

🐾 OpenClaw Full Stack — Complete Replication Guide

By Dr Rushindra Sinha (Rushi) *As of February 2026 — battle-tested on macOS (Mac mini, Apple Silicon)*

> Copy this entire list to replicate the exact setup from scratch on a fresh OpenClaw install.

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1. 📦 Node.js Global Packages (npm)

```
npm install -g openclaw          # Core gateway
npm install -g @anthropic-ai/claude-code  # Claude Code CLI (coding agent)
npm install -g clawvault         # Memory vault CLI
npm install -g clawdhub          # Skill hub CLI
npm install -g mcporter          # MCP server manager
npm install -g agent-browser     # Browser control
npm install -g clawdbot          # Bot utilities
```

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2. 🍺 Homebrew Packages (macOS)

```
# Core tools
brew install node@22             # Node.js runtime (must be v22+)
brew install python@3.14         # Python runtime
brew install ffmpeg              # Video/audio processing
brew install pandoc              # Document conversion (MD → PDF)
brew install jq                  # JSON processor
brew install gh                  # GitHub CLI
brew install bore                # Tunnel for remote access
brew install yt-dlp              # YouTube/video downloader

# AI / Media
brew install openai-whisper      # Local speech-to-text (Whisper base model)

# Audio
brew install lame                # MP3 encoding
brew install opus                # Audio codec

# Optional but useful
brew install imagemagick         # Image manipulation
```

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3. 🐍 Python venv Packages

```
# Create venv in your agent workspace
python3 -m venv ~/clawd/.venv
source ~/clawd/.venv/bin/activate

pip install anthropic             # Anthropic SDK
pip install google-api-python-client  # Google APIs (Drive, YouTube, Sheets)
pip install google-auth           # Google OAuth
pip install google-auth-oauthlib  # Google OAuth flow
pip install google-auth-httplib2  # Google HTTP auth

pip install youtube-transcript-api  # YouTube transcript fetching
pip install yt-dlp                # YouTube download (Python API)

pip install pillow                # Image processing
pip install python-pptx           # PowerPoint editing
pip install weasyprint            # HTML → PDF conversion
pip install PyPDF2                # PDF reading
pip install reportlab             # PDF generation

pip install requests              # HTTP requests
pip install beautifulsoup4        # HTML parsing
pip install lxml                  # XML/HTML parser

pip install deep-translator       # Translation (EN → HI etc)
pip install instaloader           # Instagram media download
pip install snscrape              # Social media scraping

pip install playwright            # Browser automation (Python)
pip install pydantic              # Data validation

pip install xlsxwriter            # Excel file creation
pip install webencodings          # Text encoding utils
```

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4. 🧰 OpenClaw Skills (install via ClawHub)

```
clawhub install 1password         # 1Password CLI integration
clawhub install apple-notes       # Apple Notes (memo CLI)
clawhub install apple-reminders   # Apple Reminders (remindctl CLI)
clawhub install blogwatcher       # RSS/blog monitoring
clawhub install clawhub           # ClawHub CLI manager
clawhub install coding-agent      # Delegate to Claude Code / Codex
clawhub install gh-issues         # GitHub issues + PR automation
clawhub install gifgrep           # GIF search (Tenor/GIPHY)
clawhub install github            # GitHub CLI wrapper
clawhub install gog               # Google Workspace (Gmail, Drive, Calendar)
clawhub install goplaces          # Google Places API
clawhub install healthcheck       # System security audit
clawhub install himalaya          # Email via IMAP/SMTP CLI
clawhub install mcporter          # MCP server control
clawhub install model-usage       # CodexBar usage/cost reporting
clawhub install nano-banana-pro   # Image generation (Gemini, up to 4K)
clawhub install nano-pdf          # PDF editing via natural language
clawhub install notion            # Notion API
clawhub install openai-whisper    # Local Whisper transcription (no API key)
clawhub install openai-whisper-api # Whisper via OpenAI API
clawhub install peekaboo          # macOS UI capture + automation
clawhub install sag               # ElevenLabs TTS (say-style UX)
clawhub install session-logs      # Search/analyze own session logs
clawhub install skill-creator     # Create/update custom skills
clawhub install summarize         # URL/podcast/video summarizer
```

```
clawhub install tmux              # Remote tmux control
clawhub install video-frames      # Extract frames/clips from video (ffmpeg)
clawhub install voice-call        # Voice calls via OpenClaw
clawhub install wacli             # WhatsApp CLI tools
clawhub install weather           # Weather (wttr.in / Open-Meteo)
```

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5. 🔑 API Keys Required

Add all of these to `~/.openclaw/openclaw.json` under the `env` section:

```
{
  "env": {
    "ANTHROPIC_API_KEY": "sk-ant-...",          // Required — primary AI
    "ELEVENLABS_API_KEY": "sk_...",             // Voice synthesis (TTS)
    "GEMINI_API_KEY": "AIza...",                // Image generation (nano-banana-pro)
    "PEXELS_API_KEY": "...",                    // Stock footage (backup)
    "OPENAI_API_KEY": "sk-proj-...",            // Whisper API + GPT models
    "NOTION_API_KEY": "ntn_...",                // Notion integration
    "GOOGLE_PLACES_API_KEY": "AIza...",         // Google Places search
    "CONVEX_DEPLOY_KEY": "..."                  // Convex (if using Mission Control)
  }
}
```

Google OAuth (Drive, YouTube, Sheets, Gmail)

```
# Run Google OAuth flow via gog skill or manually:
# Token files live at:
~/.config/ares/google/token.json          # Drive / Sheets / Docs / Gmail
~/.config/ares/google/youtube_token.json  # YouTube uploads only
```

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6. 🎙️ ElevenLabs Voices (pre-configured)

Character	Voice Name	Voice ID
Esports Now VO	George	`JBFqnCBsd6RMkjVDRZzb`
Haksay the Husky	Viraj (Epic/Dramatic)	`CN3wrf4wja2qO5yPhs0E`

Voice settings for Haksay:

```
{
  "model_id": "eleven_multilingual_v2",
  "voice_settings": {
    "stability": 0.25,
    "similarity_boost": 0.85,
    "style": 0.8,
    "use_speaker_boost": true
  }
}
```

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7. ⚙️ LaunchD Agents (background services)

These run automatically on startup:

Agent Label	Purpose
`ai.openclaw.gateway`	Core OpenClaw gateway (always running)
`ai.openclaw.gateway-watchdog`	Restarts gateway if it crashes
`ai.ares.dashboard`	Usage/cost dashboard at localhost:3737
`ai.ares.missioncontrol`	Mission Control UI at localhost:3001
`ai.ares.bore`	Bore tunnels for remote access
`ai.ares.killswitch`	Emergency kill switch watchdog
`com.esportsnow.youtube-research`	Daily research engine (7am)

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8. ⚠️ Known Limitations & Gotchas

Issue	Notes
ffmpeg no drawtext	Homebrew ffmpeg 8.0.1 doesn't include libfreetype. Use Pillow for text-on-image, CapCut/Premiere for text overlays.
Whisper → Urdu for Hindi	Whisper confuses Hindi and Urdu scripts. Use timing data from Whisper, write Devanagari SRT manually.
YouTube daily upload cap	`uploadLimitExceeded` = daily limit hit. Wait 24h. Separate from API quota.
WhatsApp 16MB limit	All files >16MB must go via Google Drive link.
OpenAI key in Keychain	Stored as `openclaw-openai` service in macOS Keychain. Can disappear after gateway restarts — re-add if missing.
Drive tokens expire	Always refresh with `creds.refresh(Request())` inline — never hardcode tokens in shell scripts.
Twitter/X	Always use `browser` tool. `web_fetch` is blocked by X.
Chokidar on large dirs	Never use FSEvents chokidar on dirs with 1000+ files. Use polling mode (60s interval) + debounce.
MOV files on Drive	moov atom often at end of file — stream from Drive URL, don't try to download partially.
ElevenLabs free tier	Blocked on server IPs. Must be Pro account.
python3	Always use `~/clawd/.venv/bin/python` — never system python3. Packages are venv-only.

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9. 🗂️ File System Layout

```
~/.openclaw/                  # Runtime (managed by OpenClaw)
│   openclaw.json             # ⭐ Main config
│   media/                    # Send/receive media (WhatsApp limit: 16MB)
│   media/inbound/            # Incoming media files
│   agents/main/              # Agent sessions
│   credentials/              # API key files (nanobanana.key etc)

~/clawd/                      # Ares workspace + shared scripts
│   SOUL.md                   # Ares personality
│   scripts/                  # Pipeline scripts
│   │   esports_now.py        # Esports Now full pipeline
```

```
│   │   youtube_upload.py       # YouTube upload module
│   │   youtube_research.py     # Daily research engine
│   .venv/                      # Python virtual environment
│   ares-dashboard/             # Usage dashboard (Node/Express)
│   logs/                       # Agent logs

~/sam/                          # Sam workspace (content agent)
│   SOUL.md                     # Sam personality
│   AGENTS.md                   # Workflow protocols
│   MEMORY.md                   # Long-term memory
│   memory/                     # Daily session logs
│   drafts/                     # Esports Now draft JSONs
│   scripts/                    # Content scripts
│   channels/                   # Channel bibles (Haksay, etc)
│   logs/                       # Upload logs
│   research/                   # Daily research outputs
```

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10. 🚀 Fresh Install — Exact Order

1. Install Node 22 (`brew install node@22`)
2. Install OpenClaw (`npm install -g openclaw`)
3. Run `openclaw onboard --install-daemon`
4. Add API keys to `~/.openclaw/openclaw.json`
5. Pair WhatsApp (scan QR via `openclaw gateway status`)
6. Create Python venv + install packages (`pip install -r requirements.txt`)
7. Install skills (`clawhub install <skill>` for each above)
8. Install Homebrew tools (`brew install ffmpeg pandoc yt-dlp gh bore jq openai-whisper`)
9. Set up Google OAuth (run gog skill flow)
10. Configure LaunchD agents for background services
11. Write your `SOUL.md` — this is the most important step

Generated by Sam (OpenClaw content agent) — February 2026